National and Kapodistrian University of Athens
School of Science
Department of Informatics and Telecommunications
Course Instructor: Paraskevi Roupa
Spring Semester: 2023-24

COMPUTATIONAL GEOMETRY
Computational Assignment

June 4, 2024

Implementation A: *Convex hull.*

1. Implement the following algorithms for finding the convex hull in the plane:
   - Incremental algorithm
   - Gift wrapping algorithm
   - Divide and Conquer algorithm
   - QuickHull algorithm.
2. Consider 100 random points in the plane in general position. Find the convex hull of these points using the above algorithms. Is there any difference in the results?
3. Assume that some of the given points (as many as you like) are collinear. How are the above algorithms affected (explain in detail)?
4. Visualize the steps of one of the above algorithms for the construction of the convex hull.
5. Implement an algorithm of your choice for finding the convex hull in 3 dimensions.
6. Apply the algorithm of your choice to construct the convex hull of 85 points in the space $\mathbb{R}^3$.

Implementation B: *Linear Programming.*

1. Implement the incremental algorithm for solving a linear programming problem in the plane.
2. Apply the incremental algorithm to solve the following linear programming problem:
   - $max\{-3x_1 + 12x_2\}$
   - subject to the constraints:

$$\begin{aligned} x_1 - 2x_2 &\geq 1 \\ 2x_1 - 3x_2 &\geq 6 \\ -x_1 + 3x_2 &\leq 0 \\ -x_1 + 6x_2 &\leq 12 \\ 4x_1 - 9x_2 &\leq 27 \\ x_1, x_2 &\geq 0. \end{aligned}$$

Implementation C: *Delaunay Triangulation.*

1. Implement an algorithm of your choice for the Delaunay triangulation of a set of n points in the plane.
2. Considering various numbers of points in the plane, display the Delaunay triangulation. How does the algorithm you have chosen behave as the number of points increases?

Implementation D: *Geometric search.*

1. Let P be a set of n points in the plane. Implement an algorithm that constructs the corresponding *kd*-tree for P.
2. Implement an orthogonal range search algorithm based on the *kd*-tree for the point set P.
3. Consider a set P of 100 points in the plane and an orthogonal range. Applying the algorithm you have implemented, answer the question of which points of P lie inside the orthogonal range (graphically and as a list of points).

Implementation of the algorithms

The algorithms can be implemented either in Python or using the CGAL library of C++.

Submission Instructions

Submit to the "Assignments" section of e-class a compressed file, renamed as "FullName_StudentID".
This compressed file must include:

- all the C++ or Python programs
- a text file with a detailed presentation and discussion of the results
- a presentation file (optional).

On the first page of the text file you must write your full name and your student ID number.
The assignments will be presented via Webex on a day and time to be announced.

Submission Deadline:
July 11, 2024, 18:00